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Evidence-Based Guidelines for Writing Results Sections in Geoscientific Research

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Background and Structure of Geoscientific Research Papers

Geoscientific research papers follow the standardized IMRAD structure (Introduction, Methods, Results, and Discussion) that was widely adopted in the 1980s to help readers find specific information easily. These papers must include clear qualitative or quantitative evidence rather than just anecdotal reports, with careful attention to writing clarity since audiences span multiple disciplines. (2 sources)

The structure of geoscientific research papers has evolved to follow established academic conventions that promote clarity and reproducibility. The formal IMRAD structure (Introduction, Methods, Results, And Discussion/conclusions) was adopted in the 1980s and is now the format most widely used and encouraged by scientific journals, including those in geodynamics and related geoscientific fields (Zelst et al., 2021). This standardized approach facilitates modular reading and helps readers locate specific information in pre-established sections of an article (Zelst et al., 2021).

Within this framework, geoscientific research papers must meet specific evidence standards and communication requirements. All research articles should include qualitative and/or quantitative evidence rather than relying solely on anecdotal reporting (Hillier et al., 2021). Authors must also consider their diverse audience, as readers span many fields and disciplines within the geosciences, requiring clear and concise writing that avoids unnecessary jargon while including critical structural elements such as introduction, methods, results, discussion, and conclusions (Hillier et al., 2021). The methods and results sections are particularly critical, as reviewers will criticize incomplete or incorrect descriptions and may recommend rejection when these sections fail to make results reproducible and replicable (Zelst et al., 2021).

Evidence

(Zelst et al., 2021)

101 Geodynamic modelling: How to design, carry out, and interpret numerical studies

> *"it is important to make those results understandable and reproducible in the methods and results sections. Reviewers will criticise incomplete or incorrect method descriptions and may recommend rejection because these sections are critical in the process of making the results reproducible and replicable"*

> *"The formal IMRAD structure (i.e., Introduction, Methods, Results, And Discussion/conclusions) was adopted in the 1980s, and at present, is the format most widely used and encouraged by scientific journals. The IMRAD structure facilitates modular reading and locating of specific information in pre-established sections of an article (Kronick, 1976). In geodynamics, the general structure of a manuscript follows the IMRAD structure"*

(Hillier et al., 2021)

Editorial: Geoscience communication – Planning to make it publishable

> *"all research articles should include qualitative and/or quantitative evidence, and not solely anecdotal reporting; and"*

> *"all research articles should include an explicitly marked section that considers the ethics of the investigation and should also demonstrate how the research has received ethical clearance from their research institute or professional body"*

> *"Remember that the audience of GC spans many fields and disciplines. When writing your paper, please endeavour to write clearly and concisely, avoid jargon"*

> *"and include critical structural elements you will be familiar with (e.g. introduction, methods, results, discussion, conclusions)."*

Core Content Guidelines for Results Sections

Results sections should present findings in an orderly progression from general to specific analyses using neutral language, with quantitative arguments supporting the initial hypothesis while avoiding interpretation or references to other studies. (2 sources)

The results section serves a specific purpose in presenting research findings without interpretation or analysis. The primary function is to present quantitative arguments to support the initial hypothesis, while reserving any interpretation of results or references to other studies for the discussion section (Zelst et al., 2022). This separation ensures that readers can distinguish between observed findings and their scientific meaning.

Results should be organized systematically to guide readers through the data. The main results obtained from data analysis must be presented in an orderly manner, progressing from the most general and descriptive analyses to the most specific and complex findings (Garcia et al., 2021). This logical progression helps characterize the study and provides answers to the research objectives.

The presentation format should prioritize clarity and avoid redundancy. Results must be communicated using neutral and objective language through a combination of text, tables, and figures, with authors choosing the most appropriate and efficient format for each type of data while avoiding repetition of information (Garcia et al., 2021). Visual elements like illustrations, figures, and tables represent the most efficient way to present results, but authors should only include material and information that is relevant to demonstrate the scientific arguments that will be discussed in subsequent sections (Zelst et al., 2022).

To maintain focus and readability, authors should present additional supporting data as supplementary materials rather than cluttering the main results section with non-essential information (Zelst et al., 2022).

Evidence

(Zelst et al., 2022)

101 geodynamic modelling: how to design, interpret, and communicate numerical studies of the solid Earth

> *"the results section is to present quantitative arguments to the initial hypothesis. However, any interpretation of the results or reference to other studies should be reserved for the discussion. For example, results in a mantle convection model might show that dense material accumulates at the bottom of the domain (i.e., core-mantle boundary). The interpretation of these results is that they provide a mechanism to explain how LLSVPs (large low shear velocity provinces) have formed. Illustrations, including figures and tables, are the most efficient way to present results (Section 7). However, authors should only include material and information that is relevant to demonstrate the scientific arguments discussed in the next section. Therefore, to avoid distraction, writers should present additional data as supplementary materials, e.g., movies of the whole simulation, whereas only a few snapshots are provided in the main body of the manuscript."*

(Garcia et al., 2021)

Cómo escribir y publicar artículos científicos (II). Seguimos el viaje: resultados, discusión, y algo más

> *"In the "Results" section, the main results obtained with the data analysis are presented in an orderly manner, from the most general and descriptive analyses to the most specific and complex, seeking to characterize the study and to give answers to its objectives.*

Results are presented in neutral and objective language, using a combination of text, tables and/or figures, choosing the most appropriate and efficient format in each case and avoiding repeating the information provided."

Statistical and Methodological Reporting Requirements

Geoscientific results sections must include specific statistical details including exact sample sizes, test descriptions, covariates, assumptions, and complete parameter reporting with confidence intervals and p-values. Custom software or algorithms central to research must be made available to reviewers and preferably deposited in public repositories. (1 source)

Essential statistical reporting requirements for geoscientific results sections include:

• **Sample size documentation**: The exact sample size (n) for each experimental group or condition, given as a discrete number and unit of measurement, plus a statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly (Zhang et al., 2021)

• **Statistical test specification**: The statistical test(s) used AND whether they are one-or two-sided, with only common tests described solely by name while more complex techniques require description in the Methods section (Zhang et al., 2021)

• **Covariate and assumption reporting**: A description of all covariates tested and a description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons (Zhang et al., 2021)

• **Complete parameter description**: A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) (Zhang et al., 2021)

• **Hypothesis testing details**: For null hypothesis testing, the test statistic (e.g. F, t, r) with confidence intervals, effect sizes, degrees of freedom and P value noted (Zhang et al., 2021)

• **Software and code availability**: For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers, with strong encouragement for code deposition in community repositories such as GitHub (Zhang et al., 2021)

Paper	Fossil Type	Geological Period Of Focus	Geographic Location Of Discovery	Taxonomic Classification	Evidence For Origin/ Diversification	Impact On Timeline
Fossil evidence unveils an early Cambrian origin for Bryozoa	millimetric, erect, bilaminate, secondarily phosphatized fossil	Early Cambrian	Australia and South China	Phylum Bryozoa (moss animals)	Fossil evidence, phylogenetic analyses, molecular clock estimations	Pushed back first occurrence by ~35 million years.

Evidence

(Zhang et al., 2021)

Fossil evidence unveils an early Cambrian origin for Bryozoa

"The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly

The statistical test(s) used AND whether they are one-or two-sided Only common tests should be described solely by name; describe more complex techniques in the Methods section.

A description of all covariates tested A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)

For null hypothesis testing, the test statistic (e.g. F, t, r) with confidence intervals, effect sizes, degrees of freedom and P value noted For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information."

Data Presentation and Communication Strategies

Effective results sections in geoscientific papers combine text, tables, and figures in the most appropriate format for each data type, progressing from general to specific findings. Clear communication includes reporting uncertainty ranges and using concise, objective language while avoiding repetition across different presentation formats. (3 sources)

Successful data presentation in geoscientific results sections requires strategic choices about format and organization. Results should be presented using neutral and objective language through a combination of text, tables, and figures, choosing the most appropriate and efficient format in each case while avoiding repeating the information provided (Garcia et al., 2021). The organization should follow a logical progression from the most general and descriptive analyses to the most specific and complex findings to characterize the study and provide answers to research objectives (Garcia et al., 2021).

Visual elements serve as the primary means of efficient communication, but must be used judiciously. Illustrations, including figures and tables, represent the most efficient way to present results, though authors should only include material and information that is relevant to demonstrate the scientific arguments that will be discussed in subsequent sections (Zelst et al., 2022). To maintain focus and avoid distraction, additional supporting data should be presented as supplementary materials rather than cluttering the main manuscript (Zelst et al., 2022).

Clear communication of uncertainty enhances the credibility and usefulness of presented findings. Results should be phrased in a concise and clear way, with reporting findings in ranges helping to communicate the uncertainty associated with them effectively (Buckeridge, 2021). This approach allows readers to better understand the precision and limitations of the research findings while maintaining the objective presentation required for the results section.

Evidence

(Garcia et al., 2021)

Cómo escribir y publicar artículos científicos (II). Seguimos el viaje: resultados, discusión, y algo más

"In the "Results" section, the main results obtained with the data analysis are presented in an orderly manner, from the most general and descriptive analyses to the most specific and complex, seeking to characterize the study and to give answers to its objectives. Results are presented in neutral and objective language, using a combination of text, tables and/or figures, choosing the most appropriate and efficient format in each case and avoiding repeating the information provided."

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101 geodynamic modelling: how to design, interpret, and communicate numerical studies of the solid Earth

"the results section is to present quantitative arguments to the initial hypothesis. However, any interpretation of the results or reference to other studies should be reserved for the discussion. For example, results in a mantle convection model might show that dense material accumulates at the bottom of the domain (i.e., core-mantle boundary). The interpretation of these results is that they provide a mechanism to explain how LLSVPs (large low shear velocity provinces) have formed. Illustrations, including figures and tables, are the most efficient way to present results (Section 7). However, authors should only include material and information that

is relevant to demonstrate the scientific arguments discussed in the next section. Therefore, to avoid distraction, writers should present additional data as supplementary materials, e.g., movies of the whole simulation, whereas only a few snapshots are provided in the main body of the manuscript."

(Buckeridge, 2021)

Comment on bg-2020-460

> *"The results were phrased in a concise and clear way and reporting findings in ranges does helps to communicate the uncertainty associated with them well."*

Transparency and Evidence-Based Reporting Principles

Evidence-based reporting in geoscientific research requires systematic transparency through clear inclusion/exclusion criteria, objective assessment strategies, and documentation of all evidence sources. Ethical considerations must be explicitly addressed, and all claims should be supported by qualitative or quantitative data rather than anecdotal reporting. (2 sources)

Transparency and evidence-based reporting form the foundation of credible geoscientific research, requiring systematic approaches that minimize bias and enhance reproducibility. Research should employ clear exclusion and inclusion criteria that help mitigate researcher bias in literature selection and data interpretation, offering more reliable and accurate answers to research questions while enhancing the scientific rigor of knowledge innovation (Kang et al., 2024). This systematic approach ensures that selected papers and data align with research objectives and can be evaluated consistently across different studies.

Objective assessment strategies strengthen the reliability of reported findings. The evaluation of data and evidence should involve multiple assessors familiar with the relevant scientific knowledge, with disagreements resolved through structured discussion to reach consensus (Kang et al., 2024). This collaborative approach reduces individual bias and improves the accuracy of evidence interpretation.

Evidence-based data collection requires comprehensive documentation of all supporting materials. For subjective judgments or interpretations, researchers should collect and preserve original source texts as supporting material, ensuring that all conclusions can be traced back to their primary evidence (Kang et al., 2024). Additionally, all research articles should include qualitative and/or quantitative evidence rather than relying solely on anecdotal reporting, and must include an explicitly marked section addressing research ethics and demonstrating ethical clearance from appropriate institutional or professional bodies (Hillier et al., 2021).

Evidence

(Kang et al., 2024)

Detrital and authigenic clay minerals in shales: A review on their identification and applications

> *"This study employs a systematic literature review method, guided by three fundamental principles to ensure transparency and objectivity throughout the review process. These principles help mitigate researcher bias in literature selection, offering more reliable and accurate answers to the research questions. They also enhance the scientific rigor of knowledge innovation and ensure that the selected papers align with the research objectives"*
>
> *"The three basic principles are as follows: 1) Clear exclusion and inclusion criteria. This study established three exclusion criteria for papers: missing basic information, completely irrelevant, and irrelevant, while defining two inclusion criteria: relevant and closely relevant. Table 1 provides comprehensive information on the screening criteria and their explanation. 2) Objective assessment strategy. The selection of each paper and the collection of coding data were assessed by three assessors familiar with the basic knowledge of clay origin. In cases of disagreement among the assessors, a final judgement was made through discussion. 3) Evidence-based data collection. For subjective judgement data (e.g. The age of clay mineral), the original text of the paper was collected and used as supporting material."*

(Hillier et al., 2021)

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